

Cem Ozcelik

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PROFESSIONAL SUMMARY

Data Scientist with an Industrial Engineering background and nearly 5 years of end-to-end ML experience spanning customer analytics, recommendation systems, marketing ML, and MLOps deployment. The Industrial Engineering foundation -- operations research, MILP, cost-function optimization -- drives every model decision: what's the asymmetric cost of targeting the wrong customer segment, at the wrong margin, with the wrong offer? Applied at scale: ALS collaborative filtering + RFM behavioral segmentation on distributed PySpark (800K+ retail sessions); Item2Vec embeddings for "frequently bought together" recommendations; an 8-hour weekly overtime reduction through ML-driven supply chain forecasting; and full credit risk MLOps (XGBoost, Docker, Django) built around the same cost-matrix logic governing pricing and conversion optimization. Proficient in Python (Advanced), SQL (Advanced), PySpark, scikit-learn, XGBoost, and containerized model deployment. Demo projects: github.com/ccemozclk/Spark-Recommendation-Engine | github.com/ccemozclk/Credit-Risk

CORE COMPETENCIES

Skills: Recommendation Systems (ALS, Item2Vec), Customer Segmentation (RFM, K-Means), Marketing ML & Personalization, Supervised/Unsupervised Learning, Pricing & Optimization, Model Lifecycle Management, A/B Testing & Experiment Design, Explainable AI

Tools: Python, scikit-learn, XGBoost, LightGBM, PySpark, Databricks, SQL (Advanced), Docker, FastAPI, Django, TensorFlow, PyTorch, Streamlit, Redis, Qdrant

WORK EXPERIENCE

Anadolu Pet Akvaryum & Effeffe

Jan 2026 -- Present

Senior Database Analyst | Maltepe, Istanbul

- Scaled a custom Django web application into a **multi-model ML deployment platform** with versioned REST APIs, enabling new models to reach production in days -- direct analog to the model lifecycle management required in marketing analytics.
- Directed predictive supply chain optimization using Python-based forecasting algorithms, sustaining an **8-hour weekly reduction in operational overtime** and providing business teams with actionable ML-driven insights.
- Translated SKU-level Association Rule Mining outputs into strategic cross-sell actions, driving continuous revenue growth.
- Tech:** Python, Django, REST APIs, ML pipelines, Association Rule Mining, PostgreSQL, Docker

Anadolu Pet Akvaryum & Effeffe

Sept 2024 -- Dec 2025

IT & Data Analytics Specialist | Maltepe, Istanbul

- Architected a full-stack internal web application from scratch (Python/Django, SOLID principles) -- the deployment foundation later scaled into the ML platform.
- Engineered predictive supply chain models in Python and PL/SQL, **reducing stockouts and improving inventory turnover**.
- Conducted Route-Level Profitability Analysis with advanced SQL, defining cost-to-profit ratios that delivered a **3% improvement in logistics efficiency**.
- Tech:** Python, Django, PL/SQL, Oracle, SOLID, full-stack deployment

Birinci Automotive

Jan 2022 -- Sept 2024

Mid-Level Data Analyst | Cayirova, Kocaeli

- Built an **unsupervised clustering framework on behavioral time-series data** for anomaly detection -- the same statistical primitives (feature engineering, cluster labeling, threshold optimization) that drive customer segmentation and lifecycle modeling in marketing analytics.
- Delivered a **15% weekly increase in critical machine availability** and **15% OEE improvement** by bridging raw sensor data and operational strategy.
- Engineered ETL pipelines in MSSQL and Python to ingest high-volume manufacturing data, replacing manual reporting workflows.
- Tech:** Python, MSSQL, ETL, unsupervised clustering, time-series analysis

Junior Data Analyst | Cayirova, Kocaeli

- Conducted EDA and applied statistical analysis to workforce datasets, uncovering patterns to optimize HR policies.
- Designed automated data workflows and ingestion pipelines, **improving reporting efficiency by 15%**.
- **Tech:** Python, SQL, statistical analysis, ETL

PROJECTS

Scalable Recommendation Engine PySpark & Customer Segmentation

End-to-end personalized recommendation system on Apache Spark/Databricks. **RFM analysis + K-Means** for behavioral customer segmentation into 5 strategic tiers (Champions, At-Risk, etc.) -- the same segmentation logic used in retail banking next-best-offer models. **ALS collaborative filtering** with implicit feedback and cold-start handling for item-level personalized recommendations. Generated logically validated affinities across hundreds of thousands of user sessions at distributed scale.

PySpark, Apache Spark, Databricks, K-Means, ALS, Streamlit | github.com/ccemozclk/Spark-Recommendation-Engine

End-to-End Credit Risk Prediction MLOps & Financial Risk

XGBoost classifier on Lending Club dataset under heavy class imbalance (<15% defaults), optimized for asymmetric cost (FP = customer friction, FN = financial loss) -- the same cost structure governing pricing decisions and customer targeting ROI. Full MLOps pipeline: raw data ingestion to containerized production app. Interactive Django EDA platform for non-technical stakeholders. **63% F1-Score** on temporal holdout (no leakage).

Python, XGBoost, CatBoost, Django, Docker, scikit-learn | github.com/ccemozclk/Credit-Risk

Retail Demand Forecasting End-to-End MLOps Pipeline

~12.83% RMSPE on 6-week daily sales across 1,115 Rossmann stores with time-based validation (no leakage). Object-oriented training pipeline. FastAPI REST inference + Streamlit dashboard with dynamic confidence intervals. Surfaced actionable marketing insight: B-type stores yield lower promotion ROI vs A/C -- demonstrating how ML output drives marketing budget reallocation decisions.

Python, LightGBM, XGBoost, Optuna, FastAPI, Streamlit, Docker | github.com/ccemozclk/Rossmann-Demand-Forecasting

EDUCATION

B.Sc. Industrial Engineering — Kırıkkale University

Sept 2016 -- Feb 2021

Rigorous foundation in statistical analysis, operations research, and optimization. Designed MILP models with IBM ILOG CPLEX. Applied AHP and TOPSIS to multi-criteria decision problems -- direct prerequisite for pricing model design and customer value optimization.

CERTIFICATIONS

Associate Data Scientist Career Track (incl. PySpark, Recommendation Engines) — DataCamp

Jan -- May 2024

Data Science & Machine Learning Bootcamp (6 months, 10+ projects) — Miuul

Mar -- Sept 2022

SKILLS

Languages: Python (Advanced), SQL (Advanced), Bash

ML & Statistical: scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, h2o.ai, Recommendation Systems (ALS, Item2Vec), Customer Segmentation (RFM, K-Means), CNNs, RNNs, Transfer Learning, A/B Testing, Statistical Analysis

Big Data: PySpark, Apache Spark, Databricks **MLOps & Deployment:** Docker, FastAPI, Django, Flask, REST APIs, Git, Linux

Databases: PostgreSQL, MSSQL, Oracle, ETL Pipelines, Redis, Qdrant (Vector DB)

Advanced: Cost-Function Optimization, MILP, AHP/TOPSIS, OOP, System Architecture, AWS (EC2, S3)

Languages: Turkish (Native), English (Fluent)